Chapter B1: You and your genes

Overview

The inheritance of genetic information from each generation to the next is a fundamental idea in science; it can help us answer questions about why we look the way we do, and build a foundation for later exploration of ideas about genetic diseases, cell division and growth, and evolution.

Topic B1.1 explores basic concepts of the genome and how it affects an organism's characteristics, through ideas about DNA and genes as the units of genetic information, the link between genes and proteins, and how the interaction between genes and the environment affects how an individual looks, develops and functions. Topic B1.2 explores inheritance by considering the effects of dominant and recessive alleles, the inheritance of characteristics, the principles of inheritance of single-gene characteristics and how sex is determined.

Understanding of the genome and emerging gene technologies are at the cutting edge of science, and they promise powerful applications to benefit present and future generations. But they also present ethical issues for individuals and society. Topic B1.3 explores some of the ideas people use to make decisions about applications of gene technology including genetic testing and genetic engineering.

Learning about genes and inheritance before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- know that living things produce offspring of the same kind, but normally offspring vary and are not identical to their parents
- know that heredity is the process by which genetic information is transmitted from one generation to the next
- know that genetic information is stored in the nucleus
- understand a simple model of chromosomes, genes and DNA
- know about the part played by Watson, Crick, Wilkins and Franklin in the development of the DNA model
- know about sexual reproduction in animals, including the role of gametes and the process of fertilisation
- know about sexual and asexual reproduction in plants, including flower structures and the processes of pollination and fertilisation.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about genes and inheritance at GCSE (9–1)

B1.1 What is the genome and what does it do?

Teaching and learning narrative

All organisms contain genetic material. Genetic material contains instructions that control how cells and organisms develop and function. Most of an organism's characteristics depend on these instructions and are modified by interaction with the environment.

Genetic material in plant and animal cells is located in the nucleus, one of the main sub-cellular structures. In organisms whose cells do not have a nucleus (e.g. bacteria) the genetic material is located in the cytoplasm.

All the genetic material of a cell is the organism's genome. In most organisms the genome is packaged into chromosomes. Chromosomes are long molecules of DNA. Genes are sections of this DNA.

In the cells of plants and animals, chromosomes occur in pairs. The two chromosomes in a pair each carry the same genes. The two versions of each gene in the pair are called alleles, and can be the same or different. A different version of a gene is a genetic variant. The genotype of an organism is the combination of alleles it has for each gene; the phenotype is the characteristic that results from this combination and interaction with the environment. Genes tell a cell how to make proteins by joining together amino acids in a particular order.

Assessable learning outcomes

Learners will be required to:

1. a) explain how the nucleus and genetic material of eukaryotic cells (plants and animals) and the genetic material, including plasmids, of prokaryotic cells are related to cell functions
   b) describe how to use a light microscope to observe a variety of plant and animal cells
   PAG1
2. describe the genome as the entire genetic material of an organism
3. describe DNA as a polymer made up of nucleotides, forming two strands in a double helix
4. describe simply how the genome and its interaction with the environment influence the development of the phenotype of an organism, including the idea that most characteristics depend on instructions in the genome and are modified by interaction of the organism with its environment

   ⓘ *Learners are not expected to describe epigenetic effects*
5. explain the terms chromosome, gene, allele, variant, genotype and phenotype
6. explain the importance of amino acids in the synthesis of proteins, including the genome as instructions for the polymerisation of amino acids to make proteins

Linked learning opportunities

Practical work:

- use a microscope to look at a variety of plant and animal cells
- extract DNA from plant tissue

Specification links:

- principles of polymerisation, and DNA and proteins as examples of polymers (C4.2)

B1.1 What is the genome and what does it do?

Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*
DNA is a polymer in which the monomers are nucleotides. Each nucleotide includes one of four different bases (adenine, thymine, cytosine or guanine). **The order of bases in a genome is the genetic code. The genetic code is modelled using letters (A, T, C, G) to represent the bases (IaS3). The order (sequence) of bases in a gene is the code for protein synthesis.** **Each set of three nucleotides is the code for an amino acid. The properties of the protein that is made depend on which amino acids are present and their order.**	7. describe DNA as a polymer made from four different nucleotides, each nucleotide consisting of a common sugar and phosphate group with one of four different bases attached to the sugar *(separate science only)*
	8. explain simply how the sequence of bases in DNA codes for the proteins made in protein synthesis, including the idea that each set of three nucleotides is the code for an amino acid *(separate science only)*
	9. recall a simple description of protein synthesis, in which: • **a copy of a gene is made from messenger RNA (mRNA)** • **the mRNA travels to a ribosome in the cytoplasm** • **the ribosome joins amino acids together in an order determined by the mRNA** ⓘ ***Learners are not expected to recall details of transcription and translation*** *(separate science only)*
The order of bases in DNA can be changed if one or more nucleotides is deleted, inserted or substituted for a different nucleotide; these are mutations, and create genetic variants. **If the sequence of bases in a gene is changed by mutation, a protein made from it may function differently or not at all, though in some cases, the mutation won't have any effect.** **Some sections of DNA do not code for a protein, but they control whether particular genes are expressed, and therefore whether particular proteins are made. Thus, mutations in these sections can also affect phenotype by altering gene expression**	**10. recall that all genetic variants arise from mutations** *(separate science only)*
	11. describe how genetic variants in coding DNA may influence phenotype by altering the activity of a protein *(separate science only)*
	12. describe how genetic variants in non-coding DNA may influence phenotype by altering how genes are expressed *(separate science only)*

Linked learning opportunities
Ideas about Science: • using letters to model the genetic code (IaS3)

B1.2 How is genetic information inherited?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
During sexual reproduction, each offspring inherits two alleles of each gene; one allele from each gamete. The two alleles can be two copies of the same genetic variant (homozygous) or different variants (heterozygous). A variant can be dominant or recessive, and the combination of alleles determines what effect the gene has.	1. explain the terms gamete, homozygous, heterozygous, dominant and recessive 2. explain single gene inheritance, including dominant and recessive alleles and use of genetic diagrams 3. predict the results of single gene crosses
Genetic diagrams such as family trees and Punnett squares can be used to model and predict outcomes of the inheritance of characteristics that are determined by a single gene (IaS3). However, most characteristics depend on the instructions in multiple genes and other parts of the genome.	4. use direct proportions and simple ratios in genetic crosses M1c 5. use the concept of probability in predicting the outcome of genetic crosses M2e
Principles of inheritance of (single gene) characteristics were demonstrated in ideas developed by Gregor Mendel, using pea plants. Mendel's work illustrates how scientists develop explanations that account for data they have collected (IaS3). **Our understanding of genetics has developed greatly since Mendel did his work; we now know that most characteristics depend upon interactions between genetic variants in multiple parts of the genome. Today, scientists sequence whole genomes to investigate how genetic variants influence an organism's characteristics.**	6. recall that most phenotypic features are the result of multiple genes rather than single gene inheritance ⓘ *Learners are not expected to describe epistasis and its effects* 7. describe the development of our understanding of genetics including the work of Mendel **and the modern-day use of genome sequencing** *(separate science only)*
A human individual's sex is determined by the inheritance of genes located on sex chromosomes; specifically, genes on the Y chromosome trigger the development of testes.	8. describe sex determination in humans

Linked learning opportunities

Practical work:

- microscopy of pollen tubes on agar (nuclei visible under high power)

Ideas about Science:

- use genetic diagrams (e.g. family trees and Punnett squares) to model and predict outcomes of single gene inheritance (IaS3)
- **distinguish data from explanatory ideas in an account of Mendel's work, and explain how Mendel's explanations accounted for the data he collected (IaS3)**

B1.3 How can and should gene technology be used?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Comparing the genomes of individuals with and without a disease can help to identify alleles associated with the disease. Once identified, we can test for these alleles in adults, children, fetuses and embryos, to investigate their risk of developing certain diseases. We can also assess the risk of adults passing these alleles to their offspring (including the identification of 'carriers' of recessive alleles). Genetic testing can also help doctors to prescribe the correct drugs to a patient ('personalised medicine'), by testing for alleles that affect how drugs will work in their body. Another application of gene technology is genetic engineering, in which the genome is modified to change an organism's characteristics. Genetic engineering has been used to introduce characteristics into organisms such as bacteria and plants that are useful to humans. Gene technology could help us provide for the needs of society by improving healthcare and producing food for the growing population. But with genetic testing we must also consider how the results will be used and by whom, and the risks of false positives/negatives and miscarriage (when sampling amniotic fluid). With genetic engineering there are concerns about the spread of inserted genes to other organisms, the need for long-term studies to check for adverse reactions, and moral concerns about modifying genomes (IaS4).	1. discuss the potential importance for medicine of our increasing understanding of the human genome, including the discovery of alleles associated with diseases and the genetic testing of individuals to inform family planning and healthcare 2. describe genetic engineering as a process which involves modifying the genome of an organism to introduce desirable characteristics **3. describe the main steps in the process of genetic engineering** **including:** • **isolating and replicating the required gene(s)** • **putting the gene(s) into a vector (e.g. a plasmid)** • **using the vector to insert the gene(s) into cells** • **selecting modified cells** 4. explain some of the possible benefits and risks, including practical and ethical considerations, of using gene technology in modern agriculture and medicine

Linked learning opportunities

Specification links:
- the involvement of genetic and other risk factors in the development of diseases such as cardiovascular disease, cancer and type 2 diabetes (B2.5)
- how can we treat disease? (B2.6)

Ideas about Science:
- genetic testing and genetic engineering as applications of science that have made a positive difference to people's lives (IaS4)
- discuss risks, benefits, ethical issues and regulation associated with gene technology (IaS4)